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University of Amsterdam

Multilingual Natural Language Processing

Overview

Today's Class

- Brief introduction to natural language processing and its applications
- Overview of general NLP applications
- Course administrativa

Prerequisites

- Basic math understanding of
  - linear algebra
  - calculus
  - probability

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  - evaluation of ML models (train/dev/test split)
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  - a comprehensive introduction to ML
    - there are many aspects of deep learning that will not be covered here (in particular many of the fundamentals)
  - a comprehensive introduction to natural language processing
    - there are many more interesting approaches in NLP not related to multilingualism

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- After decades of NLP research, it turns out, it's not that easy to model formally

Human Language Understanding

- Who did not press charges against Clarkson?

example by Blunsom (2017)

n 03:27

London (CNN) — The BBC producer allegedly struck by Jeremy Clarkson will not press charges against the "Top Gear" host, his lawyer said Friday.

Clarkson, who hosted one of the most-watched television shows in the world, was dropped by the BBC Wednesday after an internal investigation by the British broadcaster found he had subjected producer Oisin Tymon "to an unprovoked physical and verbal attack."

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  - document/discourse structure

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The Relevance of NLP in Industry

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- Many companies in the IT sector invest in language technology
  - Google, Microsoft, Baidu, IBM, Huawei...
  - Amazon, Bol, Alibaba, Booking...
  - Bloomberg, Reuters, Elsevier, ...
  - ...

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- In question answering
  - users enter well-formed questions, e.g., *What is the capital of Poland?*
  - the QA system returns (a list of) actual answer, e.g., *1962* (Cuba Crisis), or *Warsaw* (Poland's capital)

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- For example: *President [Biden] has received the French prime minister [Macron]*
- Seems trivial for English (but still makes silly mistakes), but much harder for other languages

Named Entity Recognition

- The Guardian newspaper web site

He added: “Our writers have interviewed Steve Bannon for the New Yorker before, and if the opportunity presents itself I’ll interview him in a more traditionally journalistic setting as we first discussed, and not on stage.”

⋮

Bannon said in a statement: “The reason for my acceptance was simple: I would be facing one of the most fearless journalists of his generation. In what I would call a defining moment, David Remnick showed he was gutless when confronted by the howling online mob.”

- Steve Bannon → `theguardian.com/us-news/steve-bannon`
- David Remnick → `theguardian.com/books/david-remnick`

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Story highlights

Producer Oisin Tymon will not press charges against Jeremy Clarkson, his lawyer says

An internal BBC investigation found Clarkson had struck Tymon in an "unprovoked attack"

The BBC dropped Clarkson as "Top Gear" host Wednesday and police asked for the report

London (CNN) — The BBC producer allegedly struck by Jeremy Clarkson will not press charges against the "Top Gear" host, his lawyer said Friday.

Clarkson, who hosted one of the most-watched television shows in the world, was dropped by the BBC Wednesday after an internal investigation by the British broadcaster found he had subjected producer Oisin Tymon "to an unprovoked physical and verbal attack."

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- Operates on the sentence and document level

Sentiment Analysis

njay22

★★★★★ **everyone needs to get one!**

3 May 2017

Colour: Bluetooth & Wi-fi | **Verified Purchase**

loving it - cooked a medium steak! Perfect! finished it off with george foreman grill ... cant wait to explore many more recipes

highly recommend it!

4 people found this helpful

- Very positive but not very informative

Sentiment Analysis

Rhaoulos

Expensive but good

23 July 2018

Colour: Bluetooth & Wi-fi | Verified Purchase

This simple device allows you to discover sous-vide for a high price with useless wifi and bluetooth (honestly this is completely useless and it is not even compatible with Alexa).

Although it is expensive and is the only one I tried, it keeps the water temperature at a very steady temperature for hours.
This is a new way of cooking that often requires a finishing sear in the pan.
I tried it with steaks, burgers, chicken breasts, chicken legs (for chicken confit) and I reached some unexpectedly good results.

The only downside is the time it takes as it is slow (pre)cooking. But the result is a moist and tender food like you have never seen.

- Very positive but ...

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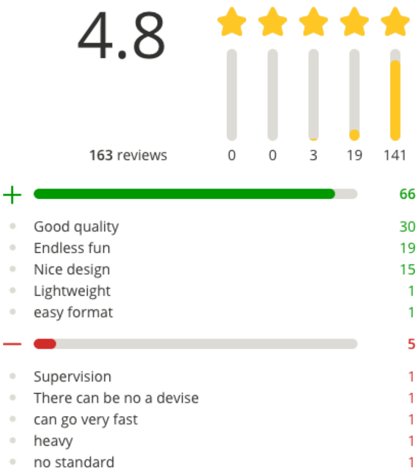

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- Requires 'understanding' the source sentence and the ability to generate sentences in the target language

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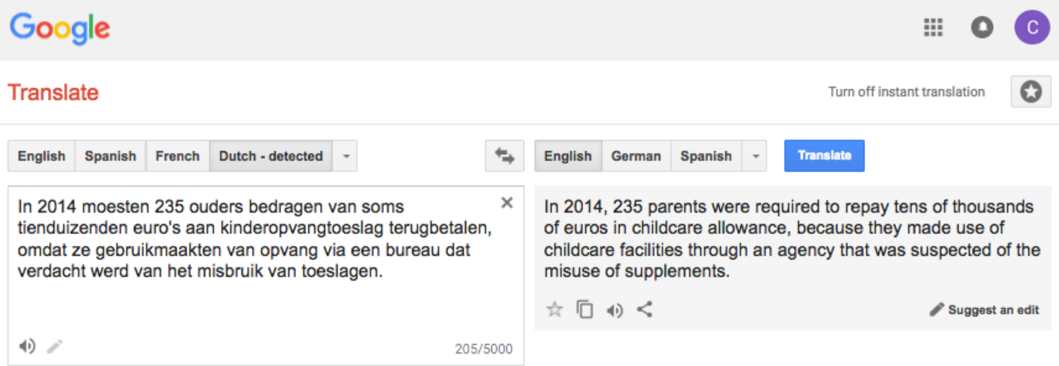

- State-of-the-art neural machine translation often achieve impressive quality

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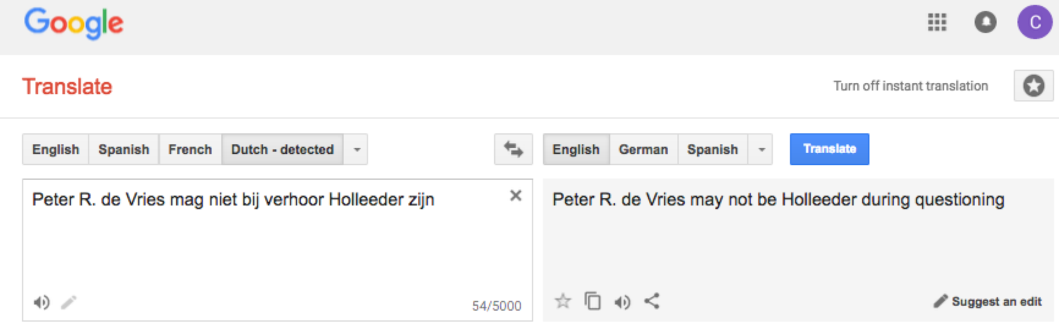

- State-of-the-art neural machine translation can also fail spectacularly

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- Machine learning is then used to weigh the importance of individual features to make correct predictions

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  - can get it spectacularly wrong at times

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  - robotics
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  - mimic human dialog interaction

LLMs and Translation

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The Ethiopian embassy in Khartoum, where the Ethiopian peacekeepers were detained, is closed. [Mistral Small 3.2]

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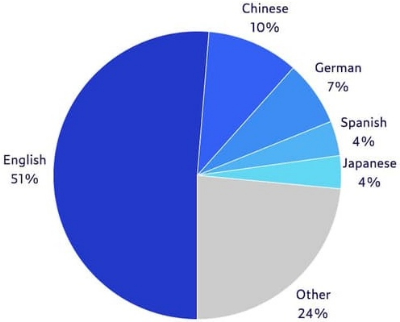

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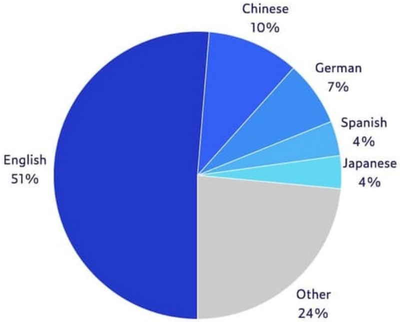

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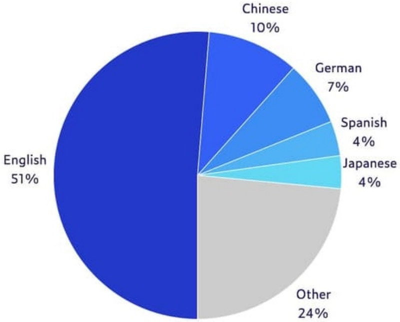

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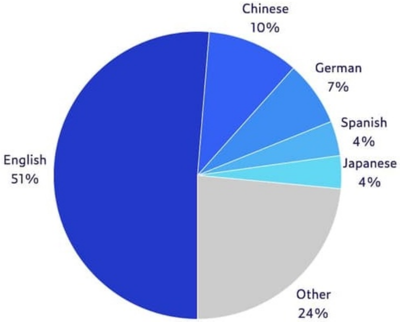

- Most state-of-the-art models are therefore English-centric
- Multilingual models (Llama 3.1, Deepseek) exhibit better performance for high-resource languages
- Dedicated multilingual models (Aya-Expanse) tend to lag behind

Multilingual Scenarios

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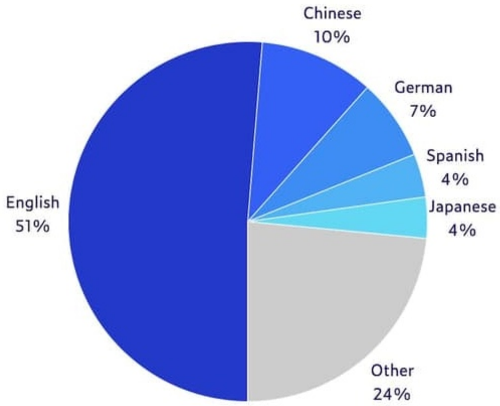
Crosslinguality



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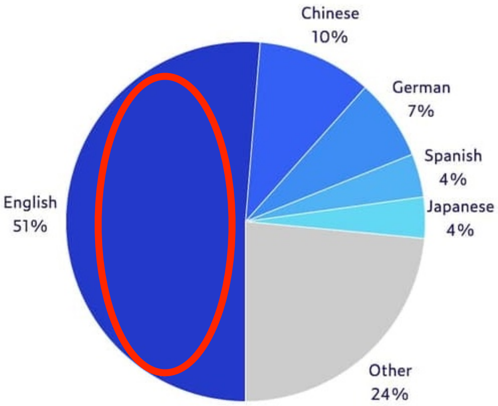

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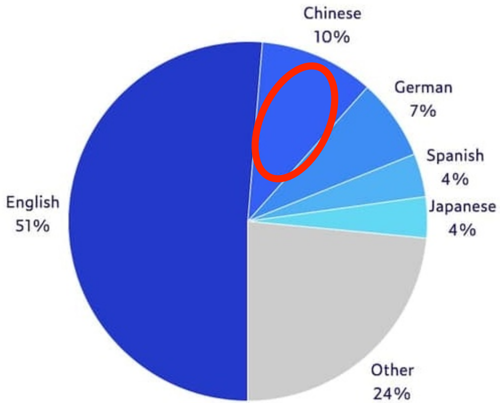
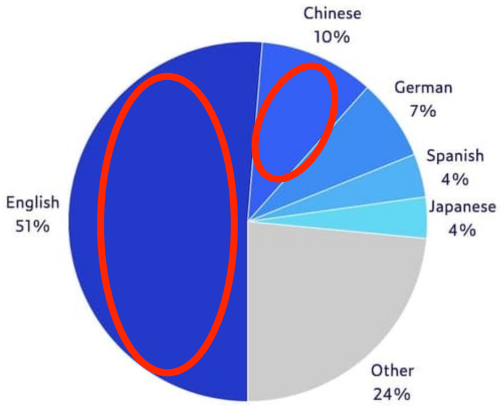
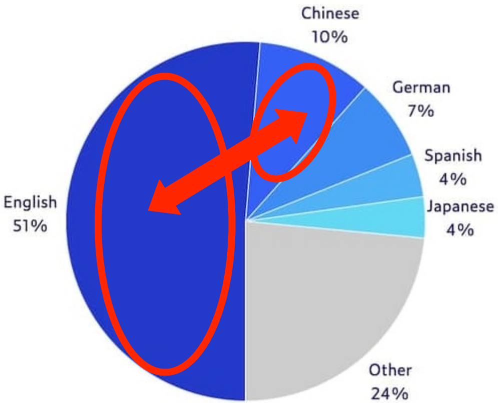


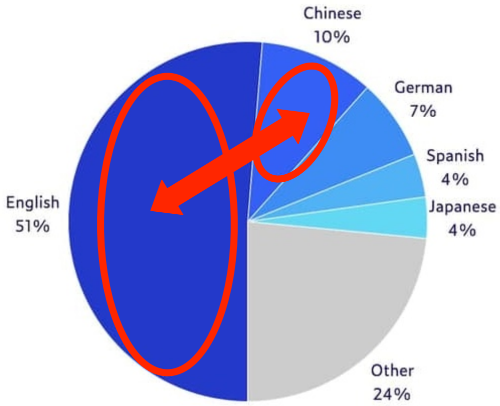
Crosslinguality



Crosslinguality



Crosslinguality



How does knowledge transfer across languages?

Crosslingual Scenarios

- Machine Translation
  - How to make information understandable across languages?

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  - How to extract information from resources in other languages?

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  - How to manipulate information across languages?
- Crosslingual Reasoning
  - How to combine information across languages?

Recap

- Brief introduction to natural language processing and its applications
- Overview of general NLP applications
- Course administrativa